

KEY CONCEPTS

Bioremediation Synthetic Biology

AI Scoring Microbe Design

Kill Switches Env. Profiling

Weighted Ranking Biosafety

PROBLEM DEFINITION

Environmental pollutants — heavy metals, hydrocarbons, dyes, pesticides — resist conventional remediation. Biological approaches depend on **pollutant type, organism suitability, pH, temperature, cost, and safety** — variables too complex for manual trial-and-error. No unified intelligent tool exists to recommend, design, and simulate engineered microbial solutions at scale.

POLLUTANTS TARGETED

Pb Lead
Sequestration / Binding

Dy Textile Dye
Enzymatic Degradation

Hc Oil / Hydrocarbons
Hydrocarbon Degradation

Ps Pesticides
Detoxification Pathway

CANDIDATE CHASSIS ORGANISMS

● *Pseudomonas putida* Hydrocarbon specialist

● *Bacillus subtilis* Robust & versatile

● *E. coli* (lab chassis) Genetic toolkit model

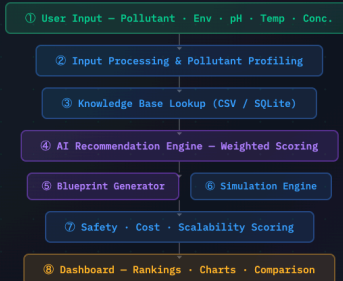
● *Microalgae* Photosynthetic biosorption

● *Microbial Consortium* Synergistic multi-species

PROPOSED SOLUTION

SynBioRem AI is an intelligent recommendation platform that ingests a user-defined pollution scenario and outputs **ranked synthetic microbe designs** with mechanism blueprints, safety evaluations, and deployment strategies. Users interactively tune pH, temperature, and concentration to observe real-time changes in remediation efficiency through a dynamic simulation layer.

SYSTEM ARCHITECTURE — FLOWCHART



SCORING FORMULA

Final Score =

- 0.35 × Remediation Match
- + 0.20 × Environment Compatibility
- + 0.20 × Safety Score
- + 0.10 × Cost Efficiency
- + 0.15 × Scalability Index

HOW IT WORKS

- User defines a **pollution scenario** via interactive sliders and dropdowns
- Pollutant profiler classifies the input and determines the required **remediation mechanism**
- Knowledge base supplies organism–pollutant mappings, environmental ranges, and safety data
- AI engine applies **rule-based filtering + weighted multi-criteria scoring** to rank candidates
- Blueprint generator outputs a synthetic design card: chassis, pathway, kill switch, deployment
- Simulation layer models **concentration reduction over time** as users adjust parameters

EXAMPLE OUTPUT — SCORING

Scenario: Oil · Soil · pH 7 · 30°C

Remediation	0.92
Env. Compat.	0.88
Safety	0.78
Cost	0.82
Scalability	0.85

Top Pick → *Pseudomonas putida*

BLUEPRINT CARD (SAMPLE)

Chassis: **P. putida**

Mechanism: **Hydrocarbon degradation**

Safety: Nutrient-dependency kill switch

Deploy: Open soil / semi-contained bed

Suitability: ● High

SOFTWARE STACK

REFERENCES

Singh et al., 2020 — Microbial remediation of pollutants · Chen & Wang, 2021 — Synthetic biology for bioremediation · IGEM Foundation — Kill switch design standards · EPA — Bioremediation guidelines

CONTACT

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